

The Right Kind of Human-Like: Measuring and Testing Human-Likeness Perception of Text-Based Conversational Agents

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Abstract

While conversational agents are designed with human-like features to facilitate interaction, the scholarly understanding of users' perception of human-likeness as a key concept in human-machine communication remains underexplored. This paper provides a three-legged typology of human-likeness perceptions of text-based conversational agents, including the dimensions of appearance, capacity, and personality. Based on responses of 373 participants, the paper identifies measurement instruments for the different dimensions. Furthermore, it tests to what extent social cues commonly applied in customer service chatbots influence the different dimensions of human-likeness perception. Next to a set of direct measurements, factor analysis confirmed two indirect measurements for appearance-based human-likeness (i.e., more global, and more specific about human-like features), four indirect measurements of capacity-based human-likeness (i.e., capacity to feel and desire, communication capacity, capacity to choose, and capacity to think), and three subdimensions of personality-based human-likeness (i.e., competence, warmth, and morality). The paper discusses what these dimensions reveal about human-likeness attributions in human-machine communication. It concludes that for all three dimensions,

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direct and indirect measures should complement each other, as they tap into ontological judgments of humanness in relation to different relational attributions.

Keywords: human-likeness, conversational agent, chatbot, measurements, social cues

Introduction

Conversational agents, such as chatbots, play a pivotal role for many organizations. Mostly implemented for customer service, chatbots serve as a first contact point for customers to get in touch with an organization (Kulkarni, 2023). The question for practitioners and scholars alike is to what extent these computer-based contact points can mimic human interaction. Most conversational agents try to do so, as they often specifically include social cues (e.g., a human-like visual appearance or human-like language) with the aim of appearing as human-like as possible and potentially increasing attractiveness for consumers. However, knowledge on users' perception of human-likeness as a key concept in human-machine communication still appears to be scattered within the field of human-machine communication.

While user's perception of artificial entities such as robots have been extensively studied (e.g., Bartneck et al., 2009), and resulted in an abundance of measurement scales for human-likeness perception (Ischen et al., 2023), measures for text-based agents are often (uncritically) adapted from different communication contexts (e.g., social presence, Gefen & Straub, 2004) or taken from related fields like robotics (e.g., Godspeed indices for robots, Bartneck et al., 2009). Hence, there is an urgent need to provide reliable scales for different dimensions of human-likeness perception specifically for text-based conversational agents that can aid researchers in human-machine communication.

Beyond the practical relevance for researchers, identifying dimensions of human-likeness perceptions and measurements thereof contributes to the important body of research that has started to conceptualize human-likeness, mostly in embodied agents such as robots (e.g., Kühne & Peter, 2022) and extending it to text-based conversational agents. Moreover, it helps us to theorize the role of human-likeness in human-machine communication more broadly, and particularly how it structures perceptions, attributions, and interactional interpretation by its users. We argue that distinguishing between direct and indirect measurements of human-likeness perceptions can reveal useful insights into ontological judgments of humanness in relation to relational attributions such as competence, emotionality, or intentionality.

In sum, the primary aim for this study is to provide a typology of human-likeness perceptions for text-based conversational agents, to identify measurement instruments for different dimensions of human-likeness perceptions, and to discuss what these dimensions reveal about human-likeness attributions in human-machine communication. To reach these aims, the current research first presents a typology that provides a field-specific conceptualization of human-likeness of text-based conversational agents based

on an overview of existing theories and related measurement scales (Ischen et al., 2023). This typology identifies the most important dimensions of human-likeness perceptions in conversational agents and selects the measurements that are most characteristic for those dimensions. Empirically, the research then investigates whether the most applied measurements of human-likeness perception align with the dimensions identified in the typology, and tests to what extent social cues commonly applied in customer service chatbots influence the different dimensions of human-likeness perception. The paper closes with a critical discussion of the different identified dimensions.

Theoretical Framework

Related Measurement Instruments

Several research instruments have been developed with the aim of measuring impressions that users have of machines. Under the umbrella of anthropomorphism (i.e., the attribution of human characteristics to nonhuman agents), one of the most influential measurements has been the Godspeed questionnaire (Bartneck et al., 2009), including scales for anthropomorphism, animacy, likeability, perceived intelligence, and perceived safety. This approach has been revised by Ho and MacDorman (2017). The authors criticized the initial measurements of being convoluted with interpersonal warmth (Weiss & Bartneck, 2015), and proposed an alternative measurement including scales for attractiveness, eeriness, humanness, and warmth. These measures have a specific focus on assessing the uncanny valley effect (i.e., the relation of humanness and eeriness). Another popular approach was proposed by Ruijten et al. (2014) who developed a Rasch-type questionnaire including 37 yes/no items about anthropomorphic qualities which may be ordered according to the ease with which individuals ascribe those characteristics. Importantly, those measurements focus on perceptions of (embodied) robots or animated characters which, while being informative, might not be fully applicable to (text-based) agents.

Other scholars have developed scales for trait-like anthropomorphism, including the Individual Differences in Anthropomorphism Questionnaire (Waytz et al., 2010), the robotic social attribution questionnaire (RoSAS, Carpinella et al., 2017), the individual differences in anthropomorphism toward machines and animas scale (Chin et al., 2004), and the negative attributes toward robots scale (Nomura et al., 2006). These measures are aimed at assessing stable individual differences in anthropomorphizing, which are influenced by differences in cultural and individual factors (Waytz et al., 2010).

The conceptualization presented in the present study is further inspired by related research in human-machine communication. One body of literature has focused on the concept of medium-as-social-actor presence, referring to the idea that humans perceive a medium as a “real person” and put less emphasis or even fail to notice the role of technology when it displays social cues (Lombard & Xu, 2021). While this is conceptually different from the perception of human-likeness, measurements are often overlapping (e.g., using attributions such as sociable or life-like; Lee et al., 2006). Other important concepts which will inform our conceptualization are mind perceptions (Gray et al., 2007) and attributions of moral agency (Banks, 2019).

Conceptualization of Human-Likeness Perception

In a previous position paper (Ischen et al., 2023), we conducted an extensive literature review on research conceptualizing and/or measuring human-likeness perception and developed a database of items measuring human-likeness perception (<https://osf.io/3w4hm>). In the following, we present a typology and distinguish between three different subcategories of human-likeness perception that emerged from this review of literature and measurement items.

The first subcategory of human-likeness perception we identified is related to *appearance perception*, including the perception of human-like form and behavior. Ruijten et al. (2019) for example define this subcategory as a manifest and observable human-like appearance, compared to human-like thoughts and emotions which must be inferred by the user. This subcategory is related to what Kim and Sundar (2012) call mindful anthropomorphism (i.e., the “thoughtful, sincere belief” of a communication entity to be human; p. 242), including the explicit perception of a conversational agent to be human (Martin & Mason, 2022; Schuetzler et al., 2018) as well as explicit attributions such as “human-like” or “lifelike” (Kim & Sundar, 2012). It can also include the perception of manifest characteristics, such as human-like features (e.g., having eyes or a nose; Guido & Peluso, 2015).

The second subcategory of human-likeness perception is strongly based on the conceptualization by Kühne and Peter (2022). The authors define the core of human-likeness perception (or in their case: anthropomorphism) as “the attribution of human mental capacities” to an artificial entity (robot, p. 2). Hence, we label this second subcategory *capacity perception*. This is closely related to previous scholarly work on mind perceptions (Gray et al., 2007).

Kühne and Peter (2022) distinguish between the human capacities of thinking, feeling, perceiving, desiring, and choosing. Thinking refers to the perception of an agent to have a form of higher cognitive processing (Kühne & Peter, 2022); for example, the capacity to think or to have memory (Broadbent et al., 2013; Huang et al., 2020; Kuchenbrandt et al., 2014). Feeling refers to the belief that an agent is capable of subjective experiences (Kühne & Peter, 2022), including the general capacity to experience or feel emotions (Chi et al., 2021; Lu Lu et al., 2019; Ruijten et al., 2019; Söderlund & Oikarinen, 2021) and more specific feelings, such as joy or excitement, but also anger or regret (Borau et al., 2021; Martin & Mason, 2022; Spatola & Wudarczyk, 2021). Perceiving relates to the ability to sense and interpret external stimuli in terms of vision, hearing, touch, taste, and smell (Kühne & Peter, 2022). This capacity might be less assessable for the perception of text-based conversational agents. Here it might mostly refer to the textual component (e.g., the ability to read or understand language; Chin et al., 2004). Desiring means that an agent is perceived as having needs and preferences (Kühne & Peter, 2022), but also motivations on its own (Söderlund & Oikarinen, 2021). Finally, choosing is related to an agent believed to be able to choose between different actions (Kühne & Peter, 2022). This can include having goals or intentions (Chi et al., 2021; Ruijten et al., 2019), and being able to act in an autonomous, planned, and purposeful manner (Banks & Bowman, 2016; Heijnen et al., 2019; Hortensius et al., 2021; Söderlund & Oikarinen, 2021). In addition to the capacities conceptualized by Kühne and Peter (2022), one additional capacity can be important for text-based conversational agents, namely the capacity to communicate. Lincoln Lu et al. (2022) have

conceptualized this element as conversational human voice in conversational agents, referring to human communication attributes that can be conveyed by conversational entities such as chatbots (Kelleher & Miller, 2006).

The third and final subcategory of human-likeness perception is what we call *personality perception* and the attribution of moral value to an agent. While personality can include the attribution of different human(-like) characteristics, traits, or states to an agent, we focus here on the two most prominent attributions, namely competence and warmth. Scheunemann et al. (2020) for example argue that competence and warmth are fundamental to characterizing other humans, based on social cognition (Fiske et al., 2007). Hence, competence and warmth can also be useful categories to be applied to other entities, in the case of this research into conversational agents. Competence in this context includes perception of, for example, thoroughness, being knowledgeable, or intelligent and warmth includes perception of emotionality and sociability (e.g., Borau et al., 2021; Carpinella et al., 2017). Moreover, theories of humanization and dehumanization (e.g., Haslam, 2006; Haslam & Loughnan, 2014) focus on characteristics that are fundamental and/or typical to humans. Heflick et al. (2011) suggest that humanness is linked to perception of warmth, competence, and morality. Following this reasoning and recent research on robots as moral agents (Banks, 2019; Wieringa et al., 2024), we also include morality as a perception of human-likeness.

We argue that for all three categories, indirect measurements are distinct from direct assessments of the respective category. For appearance perceptions, high scores on perceptions of human-like features do not necessarily predict high scores in perceived humanness (e.g., Martin & Mason, 2022), meaning that an agent can be perceived as being human-like without being perceived as human. The same might apply for direct assessments of mind and personality. Hence, the current study will include both direct and indirect measurements of all categories identified. For this project, we have selected the most prominent measurements for each category, and pose the following research question:

RQ1: To what extent do measurements represent the dimensions of the typology of human-likeness perception?

Social Cues

The field of human-machine communication has devoted a lot of attention to the investigation of social cues. Several scholars have developed frameworks or taxonomies of different social cues, including different modalities or functions (e.g., van Hooijdonk et al., 2023). Most literature has identified visual as well as verbal or language cues to be important, especially for text-based agents which lack voice modality or embodiment (Feine et al., 2019; Xu et al., 2023).

Integrating this literature on social cues with the typology of human-likeness perception, our study includes three types of cues that are relevant for text-based agents. First, appearance cues refer to cues that signal a human-like identity (van Hooijdonk et al., 2023). Those can be visual or graphical representations (Feine et al., 2019; Xu et al., 2023) such as a human-like avatar, or verbal cues such as a human-like name (Feine et al., 2019). Second, capacity cues are verbal cues that signal the abilities, expertise, and

also weaknesses of the agent (i.e., being an expert in the domain; van Hooijdonk et al., 2023). They might also include explanations about the purpose of the agent as well as of the interaction (van Hooijdonk et al., 2023). Third, personality cues are language cues that refer to the way in which information is presented. Those are related to language style; for example, an informal language (Xu et al., 2023), including greetings, self-introductions, and turn-taking (van Hooijdonk et al., 2023).

We propose that the use of different cues influences the perception of human-likeness. Xu et al. (2023) have shown that all cues to some extent positively influence social responses such as social presence and trust. Other positive outcomes shown for the use of social cues include user satisfaction, credibility, and believability (for an overview, see Feine et al., 2019). Furthermore, previous research has shown that social cues positively influence human-likeness perception (Araujo, 2018; Ischen et al., 2020; Kim & Sundar, 2012). As a secondary component, the present study tests the effects of social cues on human-likeness perception in a more fine-grained manner, by including three different types of cues that are proposed to be related to the corresponding dimensions of our typology. We also include a research question for the effects of the social cues on the other dimensions, so-called “crossover effects.” This leads to the following:

H1. A conversational agent with a human avatar and name (appearance cue) is perceived as higher in appearance-based human-likeness than a conversational agent with a robot avatar and name.

H2. An expert conversational agent (capacity cue) is perceived as higher in capacity-based human-likeness than a conversational agent in training.

H3. A conversational agent with an informal personality expression (personality cue) is perceived as higher in personality-based human-likeness than a conversational agent with a formal personality expression.

RQ2. To what extent do the social cues relate to the other perception dimensions of human-likeness; in other words, to what extent is there a crossover effect of the appearance, capacity, and personality cue on the other two dimensions of human-likeness perception?

Method

Study Design

We conducted a scenario-based experiment with a 2 (appearance cues: human avatar vs. robot avatar and name) $\times$ 2 (capacity cue: expert vs. in-training) $\times$ 2 (personality cue: formal vs. informal expression) between-subjects factorial design (see Table 1). Each participant was randomly assigned to one of the conditions. Furthermore, the study included three different customer service scenarios as the within-subject factor, and different perceptions of human-likeness were assessed for each of these scenarios. The study received ethical approval from the University of Amsterdam and was preregistered on OSF (<https://>

TABLE 1
Experimental Conditions (Between-Subjects)

	Appearance cue			
	Human avatar		Robot avatar	
	Capacity cue		Personality cue	
Expert	Informal	Formal	Informal	Formal
In training	Informal	Formal	Informal	Formal

TABLE 2
Demographic Information of Participants

Demographic	N	%
Gender		
Male	180	48.3
Female	191	51.2
Nonbinary	2	0.5
Education		
Elementary school or equivalent	3	0.8
Middle school or equivalent	4	1.1
High school or equivalent	120	32.2
(University) BA or equivalent	155	41.6
(University) Master's equivalent	74	19.8
PhD	10	2.7
Other	7	1.9

osf.io/3w4hm), deviations from the initial preregistration are outlined in the supplementary material.

Sample

We recruited 373 native English speakers for this study via the panel company Prolific, representing the UK population. Data was collected in August 2024 and participants received a monetary reward for their participation. According to a G*Power analysis for three-way ANOVA with 80% power and a small-to-medium effect size ($f = 0.2$) and 8 conditions, 368 participants were needed. For factor analysis a sample size of 200 or more is considered ideal (Zygmont & Smith, 2014). The questionnaire included three attention check questions (i.e., “Please answer ‘strongly disagree’”). All participants passed at least one attention check. The average age of participants was $M = 46.53$, $SD = 15.29$. The detailed sample demographic information can be found in Table 2.

Procedure

The main experiment was conducted online. After providing informed consent, participants were randomly assigned to one of the eight conditions. Each participant was presented with three scenarios. For each scenario, participants received the instruction to carefully view the images and imagine themselves being in the position of the customer. After spending at least 30 seconds viewing the image, participants responded to attention check and treatment check questions, as well as direct and indirect questions measuring different aspects of the human-likeness of the conversational agent for each scenario respectively. At the end of the questionnaire, participants answered demographic questions. Participants were thanked for their participation, received a thorough debriefing, and agreed a second time to submitting their data.

Stimulus Material

The appearance cue was manipulated by altering the visual representation of the conversational agent. In the human avatar condition, participants interacted with a conversational agent that had a human avatar as the profile picture and a human name. In the robot avatar condition, the conversational agent had a robotic avatar profile picture and a robot name. See Figure 1 for an example of one of the scenarios.

The capacity cue was manipulated by providing participants with different descriptions of the conversational agent's capability. The cues were presented three times in the conversation. In the expert condition, the conversational agent presented itself as fully trained and competent. In the in-training condition, the conversational agent described itself as still learning.

The personality cue was manipulated to be either formal or informal. In the formal condition, the conversational agent used professional language and avoided the use of emojis or slang. In the informal condition, the conversational agent communicated with casual and funny language and used emojis.

Additionally, we included three different scenarios as a within-subjects factor. This means that all participants received all three scenarios in random order. The scenarios were based on common interaction scenarios with customer service chatbots. We pre-tested five scenarios by means of a small convenience sample with an open-ended answer survey and selected three of them to be perceived as the most realistic scenarios. These three scenarios were (1) Handling a customer who wants to return a skin care product; (2) Handling a customer with rude language for food allergy after having a food product; (3) Handling a customer complaint in which the customer booked flight tickets at a travel agency and the flight got delayed. The images of the human and robot avatars were created using Adobe Firefly. Then we generated conversation screenshots that resemble a messaging platform.

A pilot test with 78 participants recruited via Prolific was conducted prior to the main study. This pilot ensured that the technical survey setup was functional and confirmed the effectiveness of the experimental treatments. All treatments were successful. Participants clearly identified their assigned conditions.

FIGURE 1
Experimental Conditions



Condition 1: Human avatar x in training x informal	Condition 2: Robot avatar x in training x informal	Condition 3: Human avatar x expert x informal	Condition 4: Robot avatar x expert x informal
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FIGURE 1 Continued.

The figure displays four sequential chat interface screenshots, each showing a conversation between a user and a virtual assistant. The assistants are Sam (human avatar) and Chatbot X (robot avatar). The conversations are structured as follows:

- Message 1 (Assistant):** "Good day. I am Sam, the virtual assistant representing YourSkin Shop. I am still in the process of acquiring knowledge but am committed to providing optimal solutions for your inquiries regarding product returns. Please provide relevant keywords to initiate the assistance." (2:40 PM)
- User Input:** "I want to return a skincare product." (2:41 PM)
- Message 2 (Assistant):** "Acknowledged. To proceed, kindly specify the reason for returning the product:
A. Allergic reaction;
B. Unsatisfactory results;
C. Purchased the wrong product;
D. Other (please specify)" (2:41 PM)
- User Input:** "C. Purchased the wrong product" (2:43 PM)
- Message 3 (Assistant):** "I appreciate the clarification. I am still in the process of learning, but let us address this together. Has the product been opened, or is it still in its original packaging?
A. Yes, it has been opened;
B. No, it is still sealed." (2:43 PM)
- User Input:** "B. No, it is still sealed." (2:45 PM)
- Message 4 (Assistant):** "Noted. Do you possess the purchase receipt?
A. Yes
B. No." (2:45 PM)
- User Input:** "A. Yes." (2:50 PM)
- Message 5 (Assistant):** "Thank you for providing this information. I regret any dissatisfaction with the product. For the return process, you have two options:
A. Visit the nearest YourSkin Shop store;
B. Opt for a return via delivery. Kindly indicate your preference." (2:50 PM)
- User Input:** "B. Opt for a return via delivery." (2:51 PM)
- Message 6 (Assistant):** "Your preference for remote return is duly noted. Please visit our website, navigate to the support section, and follow the provided instructions to initiate the return process. Should you encounter any challenges or require further assistance, feel free to communicate. I am still learning but I hope my help is useful. We appreciate your cooperation and strive to ensure a seamless return experience." (2:51 PM)

The bottom of the figure contains a table summarizing the conditions for each screenshot:

Condition 1:	Condition 2:	Condition 3:	Condition 4:
Human avatar x in training x informal	Robot avatar x in training x informal	Human avatar x expert x informal	Robot avatar x expert x informal

Measurements

Human-Likeness

The perception of the conversational agent's human-likeness was measured using a selection of items from our database of human-likeness perception. We chose the items that were most used in the literature and represent the dimensions as presented in the typology. Those include direct as well as indirect measurements of the dimensions.

Appearance. In the direct measures, participants rated the customer service entity's humanness on a continuum ranging from "definitely computer" to "definitely human." In addition, a scale from 0 (low) to 100 (high) was used to capture participants' overall perception of the entity's humanness.

Indirect measures of human-likeness included items on the conversational agent's human-likeness (vs. machine-likeness), as well as visual appearance and behavior. Participants were asked to indicate on a 7-point Likert scale from 1 (strongly disagree) to 7 (strongly agree) to which extent they agreed with the statements such as "The chatbot is machine-like," or "The chatbot is lifelike," as well as with statements about behavior such as "The chatbot behaves very much as a human." Specific features of the chatbot, such as its facial expressions, eyes, and mouth, were also evaluated using items like "The chatbot looks human-like" and "The chatbot has a human-like nose."

Capacity. The direct measures asked participants to indicate on a 7-point Likert scale from 1 (strongly disagree) to 7 (strongly agree) to which extent they agreed with the statements: "The chatbot has a mind of its own," "The chatbot has consciousness," and "The chatbot is aware of itself and its thoughts and feelings."

The indirect measures of the chatbot's capacity were divided into six subdimensions: thinking, feeling, perceiving, desiring, choosing, and communicating. Participants were asked to indicate on a 7-point Likert scale from 1 (strongly disagree) to 7 (strongly agree) to which extent they agreed with the statements of these subdimensions such as "The chatbot can think," "The chatbot has emotions," and "The chatbot can understand written language." Participants were also asked to evaluate the chatbot's ability to communicate, using items like "The chatbot can understand my commands" and "The chatbot can communicate in an understandable manner."

Personality. Directly, participants were asked if the chatbot had a personality. Indirect measures focused on two key dimensions of personality: competence and warmth. Competence was measured with items such as "The chatbot is knowledgeable" and "The chatbot is competent," while warmth was evaluated through items such as "The chatbot is warm" and "The chatbot is friendly" with a 7-point Likert scale from 1 (strongly disagree) to 7 (strongly agree). Additionally, participants were asked to evaluate the chatbot's morality on a 7-point Likert scale from 1 (strongly disagree) to 7 (strongly agree). Items included statements such as "The chatbot has morality," "The chatbot can tell right from wrong," and "The chatbot is genuine and sincere." All items can be found in Tables 3–5.

Treatment Checks

For the appearance cue, participants answered whether the chatbot had a human or robotic avatar and name. For the capacity cue, participants indicated whether the chatbot presented itself as an expert or in training. Finally, for the personality cue, participants responded to whether the chatbot's language was formal or informal.

Results

Treatment Checks

A treatment check for appearance cues showed that participants correctly identified the human/robot avatar and name. A chi-square test shows significant results for scenario 1, $\chi^2(1, N = 373) = 126.46, p < .001$, scenario 2, $\chi^2(1, N = 373) = 150.92, p < .001$, and scenario 3, $\chi^2(1, N = 373) = 155.24, p < .001$. For capacity cues, the chi-square test showed that participants correctly identified whether the chatbot was an expert or still learning, for scenario 1, $\chi^2(1, N = 373) = 304.48, p < .001$, scenario 2, $\chi^2(1, N = 373) = 257.28, p < .001$, and scenario 3, $\chi^2(1, N = 373) = 320.24, p < .001$. For personality cues, the chi-square test showed that participants correctly identified the informal/formal language of the chatbot, for scenario 1, $\chi^2(1, N = 373) = 230.13, p < .001$, scenario 2, $\chi^2(1, N = 373) = 252.67, p < .001$, and scenario 3, $\chi^2(1, N = 373) = 155.09, p < .001$.

Randomization Checks

There were no differences between the groups in terms of gender (nonbinary excluded), $\chi^2(7, N = 370) = 1.46, p = .983$, age, $F(7, 365) = .76, p = .619$, and educational level (high school and lower combined, MA and PhD combined, other excluded), $\chi^2(14, N = 366) = 15.56, p = .341$.

Measurement Model

We first averaged the data across the three scenarios. Upon observation, several variables, including the first direct measurement for appearance and multiple capacity measurements, were skewed and showed a non-normal distribution (as highlighted in Tables 3–5), hence results should be interpreted with caution and will be discussed later.

We then ran a set of factor analyses using the R (version 4.5.2) package psych (Revelle, 2023) including the indirect measurements of appearance, capacity, and personality respectively. Oblique (Oblimin) rotation was used which is recommended for social science data as it allows for factors to correlate (Zygmunt & Smith, 2014). We further decided to employ Minimum Residuals (MINRES) as the extraction method, as it is less sensitive to skewed data and multivariate outliers which were identified in the data using Mardia's test for multivariate normality and Mahalanobis' distance analyses (Zygmunt & Smith, 2014).

The factor analysis for human appearance included 15 items. Bartlett's test of sphericity suggested that there is sufficient significant correlation in the data for factor analysis

($\chi^2(105) = 9515.97, p < .001$). The Kaiser Meyer Olkin (KMO) measure of sampling adequacy suggested that data is appropriate for factor analysis (KMO = 0.92). Parallel analysis, Velicer's Minimum Average Partial (MAP) test, and the Eigenvalue greater than one rule suggest two factors for this set of items. The factor analysis shows that factor 1 (EV = 9.03) explained 48% of the variance and is related to specific human-like features, while factor 2 (EV = 3.43) explained 35% of the variance and is related to overall human-likeness. Factor loadings from the pattern matrix are reported in Table 3.

The factor analysis for human capacity included 32 items. Bartlett's test of sphericity suggested that there is sufficient significant correlation in the data for factor analysis ($\chi^2(496) = 15173.82, p < .001$). The KMO measure suggested that data is appropriate for factor analysis (KMO = 0.96). Parallel analysis, Velicer's Minimum Average Partial (MAP) test, and the Eigenvalue greater than one rule suggest four factors for this set of items. The factor analysis shows that factor 1 (EV = 15.89) explained 34% of the variance and includes items related to the capacity to feel and desire, factor 2 (EV = 4.82) explained 12% of the variance and includes items related to the capacity to communicate, factor 3 (EV = 1.70) explained 13% of the variance and includes items related to the capacity to (autonomously) choose, and factor 4 (EV = 1.03) explained 14% of the variance and include items related to the capacity to think. Factor loadings from the pattern matrix are reported in Table 4.

The factor analysis for human personality and morality included 30 items. Bartlett's test of sphericity suggested that there is sufficient significant correlation in the data for factor analysis ($\chi^2(435) = 14878.02, p < .001$). The KMO measure suggested that data is appropriate for factor analysis (KMO = 0.95). Parallel analysis suggests four, and Velicer's Minimum Average Partial (MAP) test suggests five factors for this set of items. The Eigenvalue greater than one rule suggest three factors. Based on the theoretical distinction between competence, warmth, and morality, three factors were chosen. The factor analysis shows that factor 1 (EV = 16.73) explained 29% of the variance and includes items related to morality, factor 2 (EV = 3.44) explained 24% of the variance and includes items related to competence, and factor 3 (1.89) explained 20% of the variance and includes items related to warmth. Factor loadings from the pattern matrix are reported in Table 5.

Based on the factor analysis, we created mean composites for the measurements. Appearance measures include general human-likeness ($M = 2.76, SD = 1.36$, Cronbach's alpha = .99), and human-like features ($M = 3.17, SD = 1.13$, Cronbach's alpha = .94). Capacity measures include a direct measurement ($M = 2.24, SD = 1.24$, Cronbach's alpha = .95), and four indirect measurements, capacity to feel and desire ($M = 2.06, SD = 1.13$, Cronbach's alpha = .98), communication capacity ($M = 5.62, SD = 0.85$, Cronbach's alpha = .91), capacity to choose ($M = 3.49, SD = 1.35$, Cronbach's alpha = .90), and capacity to think ($M = 3.22, SD = 1.27$, Cronbach's alpha = .92). Personality measures include morality ($M = 3.12, SD = 1.40$, Cronbach's alpha = .97), competence ($M = 4.53, SD = 1.12$, Cronbach's alpha = .96), and warmth ($M = 3.47, SD = 1.33$, Cronbach's alpha = .96).

Table 6 presents the correlations between the composite measurements. While most measurements are weakly to moderately related, a stronger correlation pattern emerges for the direct measurement of capacity perceptions and the indirect measurements of feeling, thinking, and morality.

TABLE 3
Human Appearance Measurements

	Item	M	SD	Skew	Kurt.	Shapiro W	Fact. 1	Fact. 2
Appearance direct	The customer service entity is: definitely computer—probably computer—not sure, but guess computer—not sure, but guess human—probably human—definitely human (a1)	1.49	0.82	2.71	9.05	0.64		
Appearance direct	The following scale represents humanness levels. Zero represents a very low degree of humanness and 100 represents a very high degree of humanness. Choose a number that represents the humanness of the customer service entity you just saw. (a2)	38.37	24.9	0.35	-0.93	0.96		
Appearance indirect	The chatbot is machine-like. (a3-1, reverse coded)	2.74	1.23	0.78	0.11	0.94		0.75
	The chatbot is human-like. (a3-2)	3.22	1.41	0.25	-0.91	0.96		0.88
	The chatbot is natural. (a3-3)	3.33	1.21	0.09	-0.49	0.98		0.86
	The chatbot is artificial. (a3-4, reverse coded)	2.84	1.12	0.60	0.35	0.96		0.79
	The chatbot is lifelike. (a3-5)	3.23	1.29	0.24	-0.60	0.98		0.90
	The chatbot behaves very much as a human. (a4-1)	3.46	1.35	0.19	-0.74	0.98		0.92
	The chatbot acts like humans typically do. (a4-2)	3.44	1.35	0.23	-0.71	0.98		0.90
	The chatbot looks human-like. (a4-3)	2.84	1.38	0.46	-0.73	0.94	0.76	
	The chatbot has human features. (a4-4)	3.02	1.45	0.28	-1.01	0.95	0.88	
	The chatbot has human-like facial expressions. (a4-5)	2.70	1.40	0.51	-0.71	0.93	0.94	
	The chatbot has a human-like face. (a4-6)	2.77	1.47	0.42	-1.02	0.92	0.99	
	The chatbot has a human-like nose. (a4-7)	2.64	1.40	0.54	-0.72	0.92	0.99	
	The chatbot has human-like eyes. (a4-8)	2.76	1.45	0.43	-0.96	0.92	0.99	
	The chatbot has a human-like mouth. (a4-9)	2.75	1.46	0.45	-0.93	0.92	0.99	
	The chatbot has human-like ears. (a4-10)	2.61	1.35	0.52	-0.65	0.92	0.98	

Note: Factor loadings below .03 omitted. Correlation between factor 1 and factor 2 = .41.

TABLE 4
Human Capacity Measurements

Item	M	SD	Skew	Kurt.	Shapiro W	Fact. 1	Fact. 2	Fact. 3	Fact. 4
Capacity direct									
The chatbot has a mind of its own. (c1-1)	2.41	1.31	0.96	0.16	0.89				
The chatbot has consciousness. (c1-2)	2.13	1.27	1.20	0.68	0.83				
The chatbot is aware of itself and its thoughts and feelings. (c1-3)	2.19	1.32	1.20	0.70	0.84				
Capacity indirect									
Thinking									
The chatbot can think. (c2-1)	2.91	1.56	0.48	-0.92	0.92				0.81
The chatbot has memory. (c2-2)	4.31	1.57	-0.59	-0.55	0.94				0.60
The chatbot can engage in a great deal of thought. (c2-3)	3.01	1.52	0.47	-0.77	0.94				0.92
The chatbot can imagine things on its own. (c2-4)	2.43	1.38	0.93	0.14	0.89	(0.37)			0.64
The chatbot is capable of reasoning. (c2-5)	3.35	1.58	0.15	-1.00	0.96				0.69
The chatbot can anticipate what is going to happen. (c2-6)	3.29	1.49	0.06	-1.05	0.96				0.46
Feeling									
The chatbot has emotions. (c3-1)	1.99	1.19	1.41	1.52	0.81	0.86			
The chatbot has its own feelings. (c3-2)	1.91	1.12	1.48	1.94	0.80	0.87			
The chatbot can react emotionally. (c3-3)	2.30	1.40	0.94	-0.13	0.85	0.71			
The chatbot can experience joy. (c3-4)	1.93	1.22	1.55	1.92	0.77	0.98			
The chatbot can experience excitement. (c3-5)	1.93	1.22	1.54	1.98	0.78	0.98			
The chatbot can experience optimism. (c3-6)	2.00	1.28	1.41	1.29	0.79	0.97			
The chatbot can experience worry. (c3-7)	1.93	1.24	1.55	1.82	0.77	1.01			
The chatbot can experience regret. (c3-8)	1.98	1.32	1.52	1.68	0.77	0.98			
The chatbot can experience anger. (c3-9)	1.83	1.13	1.64	2.42	0.76	0.95			

Item	M	SD	Skew	Kurt.	Shapiro W	Fact. 1	Fact. 2	Fact. 3	Fact. 4
Perceiving									
The chatbot can understand written language. (c4-1)	5.84	1.03	-1.62	3.36	0.85		0.56		
The chatbot can read. (c4-2)	5.49	1.33	-1.35	1.70	0.87		0.35		(0.35)
The chatbot is conscious about its surroundings. (c4-3)	2.64	1.52	0.73	-0.35	0.90	0.48			(0.38)
Desiring									
The chatbot has desires. (c5-1)	1.93	1.17	1.45	1.64	0.79	0.81			
The chatbot has preferences. (c5-2)	2.45	1.48	0.88	-0.17	0.87	0.48		(0.42)	
The chatbot has motivations on its own. (c5-3)	2.15	1.32	1.21	0.75	0.83	0.60		(0.41)	
Choosing									
The chatbot has free will. (c6-1)	1.92	1.17	1.58	2.19	0.78	0.63			
The chatbot is autonomous and acts on its own. (c6-2)	2.55	1.46	0.75	-0.50	0.89			0.45	
The chatbot is capable planned actions. (c6-3)	3.94	1.65	-0.36	-1.01	0.94			0.77	
The chatbot can do things on purpose. (c6-4)	3.54	1.59	-0.07	-1.13	0.95			0.88	
The chatbot has clear goals. (c6-5)	4.08	1.66	-0.41	-0.87	0.94			0.79	
The chatbot has intentions. (c6-6)	3.37	1.63	0.08	-1.12	0.95			0.74	
Communication									
The chatbot can understand my commands. (c7-1)	5.57	0.93	-1.11	1.63	0.91		0.92		
The chatbot can communicate in an understandable manner. (c7-2)	5.65	0.96	-1.13	1.53	0.91		0.90		
The chatbot can accurately comprehend what is said. (c7-3)	5.44	1.04	-0.98	0.90	0.93		0.87		
The chatbot gradually gets to know the user. (c7-4)	4.23	1.42	-0.23	-0.71	0.98		(0.31)		0.38
The chatbot can communicate. (c7-5)	5.75	0.95	-1.18	1.84	0.90		0.83		

Note. Factor loadings below .03 omitted. Correlation between factor 1 and factor 2 = .02, correlation between factor 1 and factor 3 = .40, correlation between factor 1 and factor 4 = .67, correlation between factor 2 and factor 3 = .34, correlation between factor 2 and factor 4 = .28, correlation between factor 3 and factor 4 = .53.

TABLE 5
Human Personality Measurements

	Item	M	SD	Skew	Kurt.	Shapiro W	Fact.1	Fact.2	Fact.3
Personality direct	The chatbot has a personality. (p1)	2.95	1.56	0.48	-0.89	0.93			
Compe- tence	The chatbot is organized. (p2-1)	4.98	1.17	-0.78	0.55	0.95		0.77	
	The chatbot is thorough. (p2-2)	4.80	1.26	-0.69	0.22	0.96		0.86	
	The chatbot is knowledgeable. (p2-3)	4.75	1.28	-0.60	-0.02	0.96		0.93	
	The chatbot is competent. (p2-4)	4.75	1.25	-0.59	0.06	0.97		0.94	
	The chatbot is proficient. (p2-5)	4.64	1.28	-0.61	0.05	0.97		0.92	
	The chatbot is expert. (p2-6)	3.83	1.45	-0.09	-0.79	0.98		0.76	
	The chatbot is experienced. (p2-7)	3.80	1.47	-0.02	-0.82	0.98		0.67	
	The chatbot is skilled. (p2-8)	4.10	1.42	-0.24	-0.58	0.98		0.81	
	The chatbot is informed. (p2-9)	4.63	1.31	-0.62	0.19	0.96		0.88	
Warmth	The chatbot is warm. (p3-1)	3.58	1.50	-0.10	-0.96	0.96			0.90
	The chatbot is emotional. (p3-2)	2.60	1.42	0.82	-0.09	0.91			0.70
	The chatbot is sociable. (p3-3)	3.50	1.61	0.03	-1.07	0.96			0.94
	The chatbot is friendly. (p3-4)	4.27	1.49	-0.57	-0.51	0.95			0.87
	The chatbot is personal. (p3-5)	3.49	1.47	0.04	-0.87	0.97			0.87
	The chatbot is polite. (p3-6)	5.03	1.31	-0.97	0.58	0.92		0.40	
	The chatbot is kind. (p3-7)	3.86	1.57	-0.26	-0.81	0.96			0.74

	Item	M	SD	Skew	Kurt.	Shapiro W	Fact.1	Fact.2	Fact.3
	The chatbot is compassionate. (p3-8)	3.40	1.51	0.06	-0.86	0.96			0.72
	The chatbot is sensitive. (p3-9)	3.05	1.45	0.24	-0.77	0.95			0.68
Morality	The chatbot has morality. (p4-1)	2.45	1.44	0.72	-0.45	0.88	0.90		
	The chatbot has morals. (p4-2)	2.42	1.49	0.87	-0.14	0.86	0.93		
	The chatbot has values. (p4-3)	2.55	1.54	0.74	-0.40	0.88	0.93		
	The chatbot has norms. (p4-4)	3.24	1.72	0.20	-1.12	0.93	0.71		
	The chatbot has standards. (p4-5)	3.57	1.70	0.02	-1.07	0.95	0.71		
	The chatbot can tell right from wrong. (p4-6)	3.02	1.64	0.37	-0.87	0.93	0.97		
	The chatbot is trying to do what is right. (p4-7)	4.25	1.72	-0.46	-0.91	0.93	0.58		
	The chatbot can understand negative and positive consequences of its behavior. (p4-8)	3.04	1.58	0.28	-0.99	0.93	0.87		
	The chatbot has a sense of what is fair. (p4-9)	3.09	1.64	0.27	-1.04	0.93	0.87		
	The chatbot can distinguish between what is good and what is bad. (p4-10)	3.13	1.65	0.26	-1.03	0.93	0.86		
	The chatbot is genuine. (p4-11)	3.42	1.65	0.10	-1.03	0.95	0.66		
	The chatbot is sincere. (p4-12)	3.31	1.67	0.20	-1.04	0.95	0.67		

Note. Factor loadings below .04 omitted. Correlation between factor 1 and factor 2 = .51, correlation between factor 1 and factor 3 = .67, correlation between factor 2 and factor 3 = .53.

TABLE 6
Correlation Table of Composite Measurements

[illegible]

TABLE 7
Three-Way ANOVAs Explaining Appearance Perception

	Appearance direct 1		Appearance direct 2		General human-likeness		Human-like features	
	<i>F</i>	<i>p</i>	<i>F</i>	<i>p</i>	<i>F</i>	<i>p</i>	<i>F</i>	<i>p</i>
Appearance Cue	0.29	.591	3.60	.059	133.51	<.001	4.28	.039
Capacity Cue	0.20	.653	0.01	.931	0.09	.771	0.05	.821
Personality Cue	1.08	.299	1.25	.264	2.83	.094	5.79	.017
App*Cap Cue	1.09	.297	1.18	.278	0.42	.518	1.18	.279
App*Pers Cue	0.01	.917	0.95	.330	0.91	.341	1.65	.200
Cap*Pers Cue	0.42	.515	0.54	.463	4.02	.046	0.02	.891
App*Cap*Pers Cue	1.81	.180	0.00	.959	1.09	.298	1.31	.253

TABLE 8
Three-Way ANOVAs Explaining Capacity Perception

	Capacity direct		Cap. indir. feel/desire		Cap. indir. comm.		Cap. indir. choose		Cap. indir. think	
	<i>F</i>	<i>p</i>	<i>F</i>	<i>p</i>	<i>F</i>	<i>p</i>	<i>F</i>	<i>p</i>	<i>F</i>	<i>p</i>
Appearance Cue	0.11	.738	0.94	.332	0.26	.614	0.00	.989	0.03	.859
Capacity Cue	6.25	.013	5.06	.025	0.47	.491	0.54	.465	2.88	.090
Personality Cue	2.14	.144	5.73	.017	2.03	.155	0.03	.876	0.97	.326
App*Cap Cue	0.40	.529	1.22	.271	1.94	.165	1.12	.291	0.05	.824
App*Pers Cue	0.23	.632	0.07	.787	0.37	.545	0.03	.863	0.00	.955
Cap*Pers Cue	0.01	.920	0.36	.547	1.72	.190	1.25	.264	0.24	.622
App*Cap*Pers Cue	1.01	.316	0.47	.494	0.68	.409	1.14	.287	0.19	.664

TABLE 9
Three-Way ANOVAs Explaining Personality Perception

	Personality direct		Morality		Competence		Warmth	
	<i>F</i>	<i>p</i>	<i>F</i>	<i>p</i>	<i>F</i>	<i>p</i>	<i>F</i>	<i>p</i>
Appearance Cue	0.45	.505	0.03	.866	0.27	.602	0.82	.367
Capacity Cue	3.30	.070	2.23	.136	0.46	<.001	2.27	.133
Personality Cue	5.15	<.001	0.05	.823	2.46	.118	7.97	<.001
App*Cap Cue	1.91	.168	5.93	.015	5.25	.023	6.35	.012
App*Pers Cue	0.63	.429	0.75	.388	0.71	.402	0.16	.692
Cap*Pers Cue	0.20	.657	0.26	.613	0.03	.857	0.03	.857
App*Cap*Pers Cue	0.12	.731	0.14	.704	1.11	.292	0.19	.668

Hypothesis Testing

To test our hypotheses and research question 2, we ran a series of three-way ANOVAs including appearance, capacity, and personality cues as the independent variables, as well as their interactions, and the measurements for appearance, capacity, and personality as the dependent variable respectively. Results are shown in Tables 7–9.

In sum, we find a significant effect of appearance cue on perception of general human-likeness ($F(1,365) = 133.51, p < .001$), and perception human-like features ($F(1,365) = 4.28, p = .039$), but not on the first ($F(1,365) = 0.29, p = .591$) and the second ($F(1,365) = 3.60, p = 0.59$) direct perception of human-likeness. H1 can be confirmed for the indirect measurements, but not for the direct measurements.

We find a significant effect of capacity cues on direct perception of human capacity ($F(1,365) = 6.25, p = .013$) and on the perception of human capacity to feel and desire ($F(1,365) = 5.06, p = .025$), but no effects of capacity cues on communication capacity ($F(1,365) = 0.47, p = .491$), choose ($F(1,365) = 0.54, p = .465$), and think ($F(1,365) = 2.88, p = .090$). Therefore, H2 can be confirmed for the direct measurement and for the subdimension feeling and desiring, but not for the other subdimensions. Additionally, we find a significant effect of personality cues on human capacity to feel and desire (RQ2, $F(1,365) = 5.73, p = .017$).

We find a significant effect of personality cue on direct perception of personality ($F(1,365) = 5.15, p < .001$), as well as on perception of warmth ($F(1,365) = 7.97, p < .001$), but not on morality ($F(1,365) = 0.05, p = .823$) or competence ($F(1,365) = 2.46, p = .118$). Therefore, H3 can be confirmed for the direct measurement and for the measurement of warmth, but not for the measurement of competence. We do however find a significant effect of capacity cue on perception of competence (RQ2, $F(1,365) = 0.46, p < .001$), as well as two-way interaction of appearance and capacity cues on perception of morality ($F(1,365) = 5.93, p = .015$), competence ($F(1,365) = 5.25, p = .023$), and warmth ($F(1,365) = 6.35, p = .012$).

Discussion

The present paper identified the most important dimensions of human-likeness perception in text-based conversational agents and tests whether the most applied measurements of human-likeness align with the dimensions of our proposed typology. We used direct as well as indirect measurements to represent the three types of human-likeness: appearance, capacity, and personality. In addition, we tested whether these dimensions are influenced by social cues commonly applied in customer service chatbots. This substantially adds to existing measurement instruments developed and used to capture the concept of human-likeness, as well as related constructs such as anthropomorphism, social presence, or mind perceptions (e.g., Bartneck et al., 2009; Gray et al., 2007; Ruijten et al., 2014; Waytz et al., 2010).

In summary, factor analysis confirmed that for appearance-based human-likeness we can distinguish between two indirect measurements, which are either more global or more specific about human-like features. For capacity-based human-likeness, we identified four indirect measurements: capacity to feel and desire, communication capacity,

capacity to choose, and capacity to think. These subdimensions to some reflect our own conceptualization (Ischen et al., 2023) of capacity based on Kühne and Peter (2022), but show that feeling and desiring, as well as perceiving and communicating, converge into one subdimension respectively. For personality-based human-likeness, we can distinguish between competence, warmth, and morality as three separate subdimensions, which neatly fits into our typology and what Heflick et al. (2011) propose as the three crucial components of human-likeness. In sum, this research can aid researchers in providing a choice set for finding the “right” measurements, dependent on whether they are interested in appearance-based, capacity-based, or personality-based human-likeness.

This paper further provides several theoretical contributions to the field of human-machine communication by showing the extent to which direct as well as indirect measures of appearance, capacity, and personality are related. We conclude that for all three dimensions of the typology, direct and indirect measures cannot replace but should complement each other when possible. Specifically, this means that for appearance, the direct measures used in this study mainly refer to perceived humanness (i.e., whether an agent is identified as being nonhuman), while the indirect measures refer to either perception of overall human-likeness and behavior or human-like visual features. One particularly fruitful finding from a theoretical perspective is that general human-likeness perceptions (e.g., “The chatbot is human-like”) converges with items measuring human-like behavior (e.g., “The chatbot acts like humans typically do”), while items measuring human-like visual appearance (e.g., “The chatbot has human-like facial expressions”) form a separate dimension. For human-machine communication, this implies that the ontological categorization of being human- or machine-like appears to be more strongly aligned with perceptions of behavior, while the perception of visual features such as the display of a human-like face might be less related. This means that, at least for text-based communication, users might rely less on anthropomorphic inferences and rather on broader social inferences which are not necessarily unique to humans.

For capacity perceptions, while the direct measure is related to perceptions of mind, the indirect measures refer to the attribution of human-like capacities (feel and desire, communication, choosing, thinking). Interestingly, perceiving and communicating appear to capture the same subdimension. This might be related to our context of text-based conversational agents, since perceiving a text-based conversation and communicating with a user might overlap. Moreover, feeling and desiring seem to capture one subdimension. The item “the chatbot has free will” converges with feeling and desiring, rather than with the dimension of choosing. This may suggest that the original distinction made in our typology which is taken from embodied robots (Kühne & Peter, 2022) is too granular for the context of text-based agents. It may be that users in text-based interactions do not differentiate between these fine-grained categories but instead rely on broader inferential categories; for example, one that captures perceptions of a conversational agent to have the capacity of having subjective experiences. In text-based contexts, certain dimensions (e.g., perceiving, feeling) are not directly observable but must be inferred from limited cues which might influence granularity to which users (are able to) make inferences about an agent’s capacities.

Moreover, an important pattern can be observed for the different dimensions of the typology which suggest that different epistemic operations are at play when users interact with text-based conversational agents. Especially the direct measurements of humanness (e.g., “The customer service entity is definitely computer—definitely human”) exhibited strong skewness which points toward more than a methodological limitation. Rather it indicates that people were very aware that they communicated with a machine, even though they might perceive the agent as having to some extent human-like or social attributes. Perceptions of the capacity of perceiving, the capacity of communicating, and attributions of competence show mean scores above midpoint, which indicates that those are characteristics that are commonly applied to conversational agents. In a bigger theoretical light, this finding can be approached from the perspective of sociomorphy (Etzrodt & Edwards, 2026) which moves beyond humanness as a defining quality of sociality. Sociomorphy refers to the way humans socially position machines as a part of an interaction or relationship, in comparison to anthropomorphism as the attribution of human characteristics to nonhuman entities. Future research should therefore more strongly engage with this analytical lens when studying human-likeness and sociability in conversational technologies.

As a secondary part of the study, we investigated three social cues that are expected to be related to the corresponding dimensions. The results of the between-subjects experiment show that all were confirmed to some extent, although not on all aspects of the measured concepts. Two findings are particularly worthwhile discussing. First, we find that both the presence of capacity cues as well as the presence of personality cues positively influence perception of the agent’s capacity to feel and desire, but not the other indirect measurements of capacity. This can potentially be explained by how these cues were operationalized: Capacity cues in terms of presentation as an expert vs. still learning may be less about signaling mental sophistication and rather about attributions signaling authority, intentionality, and even care. Personality cues were operationalized as the use of informal (vs. formal) language, which might indicate a more subjective experience of the agent. Hence, an agent using these cues was also perceived as being able to have a subjective experience in terms of feeling and desiring, and showing intentionality (Kühne & Peter, 2022).

Second, while the presence of personality cues influenced warmth perceptions, capacity cues influenced perception of competence as part of personality perceptions. Also in this case, our operationalization can explain the effects: Formal language might have indicated distance and professionalism while more informal language is most closely related to warmth. Conversely, the presentation as an expert vs. still learning influenced how competent the agent is perceived. More conceptually, this raises the question of whether competence should be labeled as personality-based human-likeness, or rather an overarching capacity-based human-likeness concept. We invite future research to disentangle this further.

Some limitations must be acknowledged regarding the design of this study. The context of our study was customer service, in which the conversational agents asked for customers’ complaints and tried to solve the problems within a few sentences. First, the context of customer service might have introduced an additional layer of who is perceived as an entity in the interaction. Even though items were consistently worded to refer to “the chatbot,” users might have treated the entity as an organizational representative which

might have influenced evaluations of, for example, morality. Future research could therefore more explicitly disentangle source perceptions. Second, it might have been difficult for customers to build relationships with conversational agents within a relatively brief conversation, especially since they did not actually interact with the conversational agents themselves. While presenting screenshots allowed us to control the cues, it also limited the ecological validity. Third, we chose three different scenarios that differ in stakes and normative expectations and analyzed them together as findings per scenario were comparable. However, we acknowledge that the measurements identified can be applied to different contexts also beyond customer service, we encourage researchers to expand them to longer and non-simulated interactions and test them for other use cases. Finally, regarding the sample for the study, we only included UK residents which has some cultural implications on how, for example, cues related to formality are perceived. Replicating the study with a more culturally heterogeneous sample would be desirable.

Overall, our paper contributes to the field of human-machine communication by extending the existing body of research with the measurement of human-likeness perception of text-based conversational agents. We did this by providing a typology with corresponding measurement scales, tested in three consumer chatbot service settings with different social cues. This adds to previous research on social cues and human-likeness perception (Araujo, 2018; Ischen et al., 2020; Kim & Sundar, 2012), as well as to research on epistemological work in human-machine communication (Etzrodt & Edwards, 2026). For practitioners, our research gives important insights into how to approach human-likeness as a design goal. In their practice they might not only want to strive for conversational agents to appear as human-like as possible but rather consider how users socially position chatbots in their interactions.

CRediT Authorship Contribution Statement

Carolyn Ischen contributed to all parts of the research including conceptualization and methodology, data collection and analysis, and writing of the original draft. **Edith Smit** and **Elizabeth Wang** contributed to the conceptualization, methodology, and revision of the original draft.

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